Title: Policy Proposal on IT Ethics - Preventing Unauthorized Use of QR Code Student IDs in Attendance and Account Verification Systems

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1.2 Executive Summary

QR coded student IDs present fast and easy attendance tracking and account verification, at the same time we see that they bring out issues of privacy and security which play out when an ID is lost, borrowed, photocopied, or used by a different person. To maintain ease of use at the same time as we reduce these risks we have put in place a policy which requires that QR codes only include a secure non-readable identifier and out which personal info may not be embedded. For any transaction which access to sensitive accounts or student info we require a second form of verification in addition to scan. Also schools will put forth a simple report process for lost or compromised IDs and must be able to immediately disable those QR credentials.

All our scanning events will be put into protected audit logs which will detail who did the scan, when it was done, and what the reason was; also access to the logs and to student data will be by role in the business. Also we will have a policy of regular training for students and staff on QR code use, the issues related to ID sharing, and reporting

issues. Thus we maintain the efficiency of the QR systems' performance while at the same time we are improving privacy protection, accountability and we are at the same time decreasing unauthorized use.

1.3 Introduction

Schools report we are seeing a growth in the use of digital tools for attendance, student records management, and access to services. A very popular example of this is the QR code student ID. Instead of the old roll calls and paper sign in sheets we see students present their QR enabled ID which the system uses to log attendance that which is automatic and almost instant. This which in turn speeds up the check in process, saves staff time, reduces paper work, and produces more accurate and easy to access records which in turn makes QR based systems a great solution for the fast paced school environment.

In the present we see that many schools which have implemented QR or ID scanning for attendance we find that which which is when the exact same QR code also functions as an access to the online portals and the library systems or other services. While a QR code may appear to be very basic what in fact is is a digital representation of a student's identity. Should a code be duplicated, recorded on a phone, passed around or used by a different person the system will log that a different person did what was reported under another student's name.

This results in false attendance reports, inaccurate data, privacy issues and at times a student is blamed for the action of another. Also what we see is that these risks are very large in the education and health fields which house very sensitive info such as names, student numbers, class enrollment, attendance info, and account details.

In 2012 Data Privacy Act personal info must be processed transparently, for legitimate purposes, and in a proportionate way (National Privacy Commission). Also students have the right to know what data is collected on them, how it will be used, and what protections are in place. This policy details for schools how to use QR code student IDs securely, fairly, and with privacy in mind.

1.4 Research and Literature Review

Review of Existing Research

QR codes are very much in use today because they do not cost much, are easy to scan, and simple to put into digital attendance and identity systems. We also see that they are easy to reproduce or to pass along. A picture of an ID, a print out, or a forward image can be used by another person to present the same code which may or may not be the actual ID owner. This issue which is reported by researchers and practitioners puts at risk data integrity, privacy, and accountability which in turn calls for more in depth verification and governance measures.

Authentication is also not a process which only identifies which account is in question; it is also to determine that the person associated with that account is who they say they are. OWASP and other security entities report that strong auth requires many forms of support (factors) as opposed to single sign on. In practice a QR code may identify a student, but we see that systems should require a second factor such as a PIN, password, or one time passcode before that person is granted access to accounts or sensitive services. This multi factor approach reduces the risk that a lost, taken, or duplicated QR ID will be used by the wrong person. Also related is the issue of data minimization which we do not put full name, address, contact info, course or account details into the QR code; instead what we do is to put in a random unique ID or secure token which the students' real personal info is kept in the school's protected systems and made available only after proper verification.

This we do a way which is through data minimization which at the same time which sees to it that if a QR image is put out there it is not wide open. Also from a legal and ethical point of view schools must put in what the law and sector standards require which is to process only what is necessary, to give out clear notice to students and parents, to put in reasonable protections, and to control access and to keep a record of what is done. All of which together help to achieve the balance between what is needed for efficiency of operation and at the same time fulfilling the duty to protect student privacy and upholding accountability.

This proposal is based on Republic Act No. 10173 the Data Privacy Act of 2012 which which puts into law the protection of personal info processed in info and com systems and that it is to be handled fairly, transparently and for legitimate purposes (National Privacy Commission). For schools that is to say we are to collect and use QR scan data only for what is related to educational services; also we may not repurpose it for other forms of monitoring without notice, consent, and appropriate safeguards. Also we see the Cybercrime Prevention Act of 2012 which addresses issues of unauthorized access and use of digital systems (Republic of the Philippines, 2012). If a student's QR ID is used by another person for which they record false attendance or access accounts we see this as an issue of security in the system as well as of accountability; thus schools must put in place technical solutions, report out clearly what happened and also have processes for account deactivation which at the same time fairly determine responsibility and which also protect students' legal and privacy issues.

The Cybercrime Act also applies to this kind of activity. Ethically, the policy aligns with the ACM Code of Ethics, which stresses that computing work should prioritize the public good (Association for Computing Machinery, 2018). In a school context, that principle means technology should serve not only practical convenience but also students' safety, fairness, and dignity. The policy is also consistent with the NIST Cybersecurity Framework 2.0, which outlines core activities—govern, identify, protect, detect, respond, and recover—that organizations should follow to manage cybersecurity risks (NIST, 2024). Applied to QR ID systems, those functions guide how institutions can assess and control risks, safeguard data, notice anomalous behavior, act when IDs are lost, and restore normal operations after incidents.

1.5 Policy Proposal

Policy Guidelines

The objective of this policy is to end all instances of unauthorized use of QR code student IDs while allowing schools to utilize their systems for taking attendance and validating their student IDs. This policy shall govern students, teachers, staff, IT personnel, system administrators, and offices conducting submission of attendance records or verifying the identity of the students. The first requirement under QR code student IDs is that these should be used solely by the school. This would cover attendance management, validation of students, and utilization of services provided to students. The school will need to disclose what information is collected from each student, reasons for collecting such information, the period of retention, and the person authorized to see this data. Moreover, the QR codes shall not have any readable personal information on them.

Personal information like names, registration numbers, courses attended, attendance records, and user accounts should all be within the confines of the school network. The use of the QR scan as the sole means of accessing any sensitive accounts should also not be allowed. In case the QR code acts as a form of account authentication, a second form of identification will be required. What I mean is that possession of a lost or duplicated QR code does not qualify as entry into any student accounts. Furthermore, the school must ensure there is an easy way of reporting lost QR IDs by students. Upon reporting the loss or theft of QR IDs, then the previous code becomes inactive and a new one issued. Access logs are also mandatory in the system but limited to purposes of audit and investigations.

Lastly, there needs to be role-based access to information contained within the QR scan and students' accounts. Implementation Strategy. In implementing this initiative, our first step will be to conduct an analysis of our current QR ID system. This analysis must be done by the school's IT department along with its data protection officer or its privacy compliance officer. The analysis needs to cover the information contained in the QR scan, those accessing it, and what happens in case of loss and duplication of QR ID. Once this analysis is completed, it will help us in updating our QR code system. Our new QR code will serve as a secure
identifier instead of the readable data that was contained in it. There needs to be an extra step added for account verification as well.

School should establish its procedures with respect to lost IDs, suspicious scans, borrowed IDs, and correction requests for attendance records. Students should also be notified about the fact that they can neither share nor lend their QR ID codes. Moreover, it would be useful if the staff were properly instructed on responding to such reports, protecting students' information, and avoiding an excess disclosure of scanning results. Monitoring will take place through a regular review of access logs, user permissions, and other reports related to the problem. Expected outcomes. Submission ID: 4329 Page 7 of 10 Madrona Case Study.pdf turnitin This policy was developed in order to have a safe and reliable QR ID code system for our students. As it can be expected, students will be much safer when they are protected from identity theft, which would otherwise happen if their QR codes were lost or used by someone else.

The school also plans to enhance its data privacy practices by minimizing the information that will be collected via the QR code and educating the students on the type of information that will be collected about them. Additionally, there are certain aspects within the policy that will help with enhancing accountability. As can be noted from the unusual cases of scanning, it is possible that log reports will be reviewed. Moreover, the new policy eliminates the element of over-bearing surveillance through restricting the use of logs for only security and investigation purposes. It can thus be seen that the school is able to use the technology of QR codes while at the same time minimizing privacy concerns.

1.6 Justification of Policy

Ethical and Legal Justification

There is a need for such a policy because the school uses the QR code technology for issuing student IDs. While currently the use is only for tracking attendance purposes, there are other issues surrounding the topic. It is very likely that someone else will impersonate the student through the use of the QR code in accessing the information and carrying out other activities for the students. Based on the utilitarian approach, the policy must provide more benefits than threats. Therefore, the QR codes policy will ensure that there is ease of scanning, together with ensuring that the safety of the student is maintained.

From a deontological perspective, the institution has a duty to ensure that any breach of privacy of the students is prevented. According to the rights approach, the students have a right to privacy, equal opportunities, and security of their data.

Our policy is consistent with the Data Privacy Act from legal perspectives, due to its elements of transparency, legitimacy, and proportionality. It means that the students will be notified of scanning, and the QR codes will be used for learning purposes only. We will collect necessary information. Furthermore, there are several best practices of cyberspace in our policy such as verification, secure logging, and role-based access control. Page 8 of 10 Submission ID: 4329 Madrona Case Study.pdf turnitin Challenges and Considerations One challenge is costs.

As for enhanced security through OTP, token protection, system logs, and auto-deactivation, we might consider the technical improvements. Concerning this, it is worth noting that the school is going to implement all those measures gradually. Firstly, the school should focus on the development of privacy policies, the report of lost IDs, the procedure of QR code deactivation, and what data can be included in QR codes. Next, the problem related to the issue of convenience should also be addressed. Specifically, there are students who claim that the additional verification process is longer than just scanning. However, it should be noted that the school does not have such a policy according to which this verification is required for general attendance scanning. The only case when this extra process becomes necessary concerns sensitive QR codes that can grant access to important accounts. The issue of staff training also deserves to be raised since employees may not be fully aware of what to do with QR scan results.

1.7 Conclusion

QR codes as a method of student ID are we see as a solution which in some ways improves speed of taking roll and identifying students' attendance. At the same time we see issues with this approach which present when the codes are lost, replicated, have a picture of them taken or given out to another party which isn't the original user that's where we see the problems.

Also the issue is made larger when these codes are used not just for attendance tracking but also for financial accounts and other private school related services.

This policy ensures the institution of what may be referred to as clear safeguards against any unauthorized use of QR code student IDs. In addition to this safeguarding, the policies include securing the use of tokens, which in turn provide the dissemination of personal information in easily readable format; implementation of second level authentication on the access to private accounts; disabling of issued but later lost QR codes; logs keeping in a secure manner; implementing of role-based access; and education for all of the above-mentioned requirements.

Also in addition to these provisions, which ensure the privacy, security, accountability and eventually proper usage of technology in general. Page 9 of 10 Submission ID: 4329 Madrona Case Study.pdf turnitin This policy needs to be passed through before any educational institution expands its application of QR ID to other more sensitive activities. And in doing so, the educational institutions can make sure of ensuring the rights of the students and prevent any misuse of their personal information in the course.

1.8 Reference

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